

SHARJAH INTL, United Arab Emirates

WMO: 411960

Lat: 25.329N

Lon: 55.517E

Elev: 111

StdP: 14.64

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	51.4	53.6	31.3	25.7	84.1	35.3	30.4	82.6	21.1	71.0	19.3	70.7	4.7	120	0.416

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.3	111.7	74.3	109.7	74.7	107.7	75.4	86.1	97.2	84.9	96.4	83.9	95.6	9.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
83.8	177.9	92.4	82.4	169.6	91.7	80.9	161.2	91.4	51.0	97.0	49.6	96.3	48.3	95.2	91.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.8	16.5	14.5	DB	45.7	116.7	3.0	1.3	43.6	117.7	41.8	118.4	40.2	119.1	38.0	120.1	
			WB	41.9	88.3	3.0	1.3	39.8	89.2	38.0	89.9	36.3	90.6	34.1	91.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.7	66.1	68.7	73.9	81.6	89.4	93.7	97.2	96.9	91.9	85.4	76.7	69.3
	DBStd	11.61	3.54	4.79	5.27	5.34	4.06	3.07	3.05	2.66	2.88	3.16	3.86	3.98
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	50	28	13	1	0	0	0	0	0	0	0	0	8
	CDD50	11921	500	524	741	947	1223	1312	1463	1453	1258	1099	801	600
	CDD65	6497	63	118	277	497	758	862	998	988	808	634	351	143
	CDH74	92403	383	862	2471	6030	11283	13759	16623	16272	12219	8148	3358	995
	CDH80	59773	44	251	1016	3324	7313	9524	12165	11816	8063	4704	1371	182
Wind	WSAvg	6.8	6.3	7.0	7.2	7.2	7.7	7.4	7.4	7.0	6.3	5.9	5.9	5.8
Precipitation	PrecAvg	3.3	0.7	1.3	0.9	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.7
	PrecMax	6.6	3.9	5.6	4.9	1.7	0.8	0.0	0.0	0.0	0.4	0.0	1.5	5.7
	PrecMin	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.1	1.0	1.6	1.3	0.5	0.2	0.0	0.0	0.0	0.1	0.0	0.4	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.1	91.7	98.4	105.5	110.9	113.2	115.0	114.6	109.8	103.7	95.2	87.7
		MCWB	63.5	64.1	65.5	67.4	70.1	71.9	73.5	73.0	73.0	70.5	67.2	66.0
	2%	DB	80.8	86.3	93.4	101.6	107.6	109.8	112.4	111.3	107.1	100.5	91.8	84.2
		MCWB	63.0	63.6	65.2	67.3	70.5	73.0	75.4	75.4	73.9	71.2	68.5	65.9
	5%	DB	78.4	82.6	89.5	98.5	105.1	107.6	109.5	109.2	104.3	98.5	89.7	82.0
		MCWB	63.0	63.6	65.0	67.1	70.7	74.5	76.4	76.1	74.8	71.9	69.3	66.3
	10%	DB	75.5	79.0	85.8	95.0	102.2	105.2	107.2	106.2	102.3	96.4	87.6	79.3
		MCWB	62.5	63.7	64.8	67.6	71.1	75.3	77.4	77.4	76.0	72.7	69.7	65.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.2	70.2	72.5	76.2	81.4	85.9	87.9	87.3	85.9	82.7	76.5	72.8
		MCDB	76.8	79.0	82.9	90.7	95.3	97.7	99.3	97.9	96.5	92.0	85.9	80.6
	2%	WB	67.2	68.4	70.7	73.7	79.2	84.2	86.2	86.0	84.5	81.1	74.7	70.5
		MCDB	75.3	77.5	80.8	87.6	93.1	95.9	97.8	96.4	94.3	90.5	84.5	78.6
	5%	WB	65.6	67.1	69.4	72.4	77.7	82.8	85.0	84.6	83.4	79.8	73.4	69.0
		MCDB	74.0	76.4	79.5	86.3	92.1	94.5	97.1	96.2	93.2	89.3	83.7	77.5
	10%	WB	64.2	65.6	68.2	71.0	76.1	81.5	83.9	83.5	82.3	78.4	72.1	67.3
		MCDB	72.7	75.1	78.4	85.2	91.3	93.8	96.4	95.7	92.2	88.3	82.6	76.2
Mean Daily Temperature Range	5% DB	MDBR	19.7	21.0	21.8	24.9	26.3	25.1	22.3	22.9	24.0	24.3	22.0	20.5
		MCDBR	23.6	25.8	28.2	30.1	30.2	28.6	25.5	25.9	26.4	26.5	24.5	23.5
		MCWBR	10.5	10.7	11.1	11.5	11.6	11.9	11.1	11.0	11.3	11.0	10.2	10.7
	5% WB	MCDBR	20.0	20.7	21.2	24.7	26.1	24.2	20.9	21.7	22.8	22.6	22.1	20.4
MCWBR		10.3	10.0	9.7	11.0	11.5	11.1	9.9	10.2	10.1	9.9	9.8	10.1	
Clear-Sky Solar Irradiance	taub	0.438	0.497	0.557	0.590	0.586	0.636	0.724	0.686	0.596	0.512	0.453	0.435	
	taud	2.094	1.922	1.779	1.731	1.739	1.680	1.564	1.634	1.769	1.928	2.082	2.104	
	Ebn at Noon	253	246	239	235	235	222	203	210	226	239	246	248	
	Edh at Noon	45	58	70	75	74	78	88	82	70	56	45	43	
All-Sky Solar Radiation	RadAvg	1271	1522	1803	2087	2342	2387	2254	2125	1965	1710	1380	1203	
	RadStd	83	122	123	126	48	44	57	47	28	25	43	79	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.99	+1.10	N/A	N/A	N/A	N/A	N/A	N/A	+263	+254
(47)	Regional (14 neighbors)	+0.85	+1.26	N/A	N/A	N/A	N/A	N/A	N/A	+270	+268

Nomenclature: See separate page